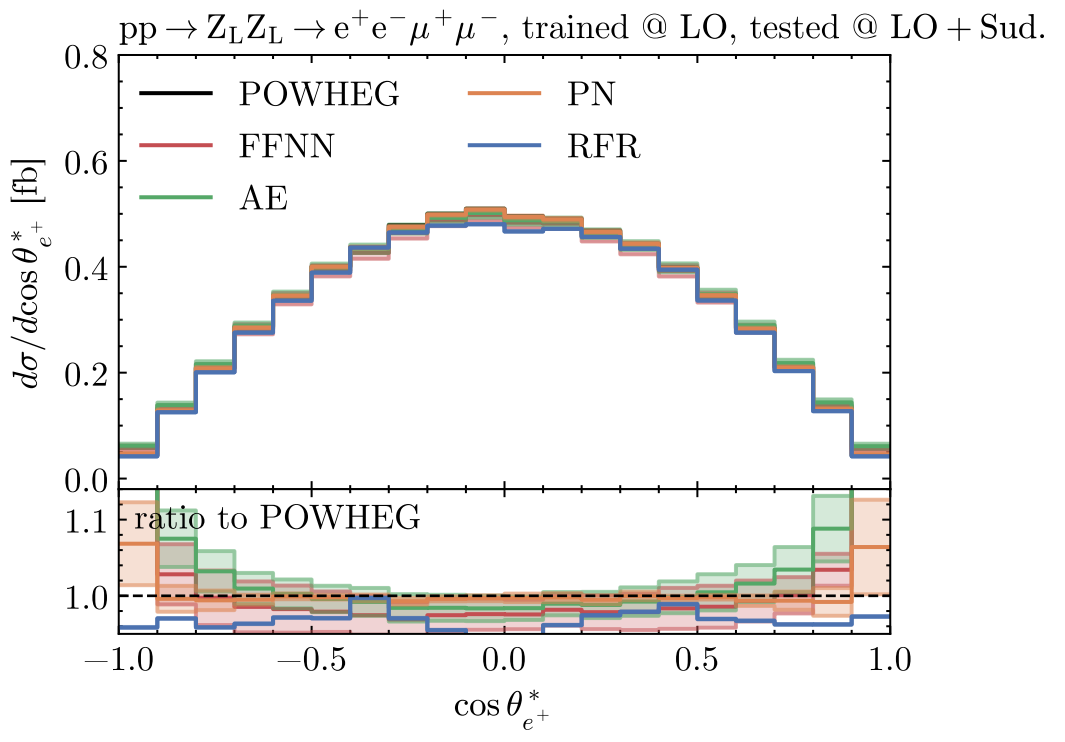
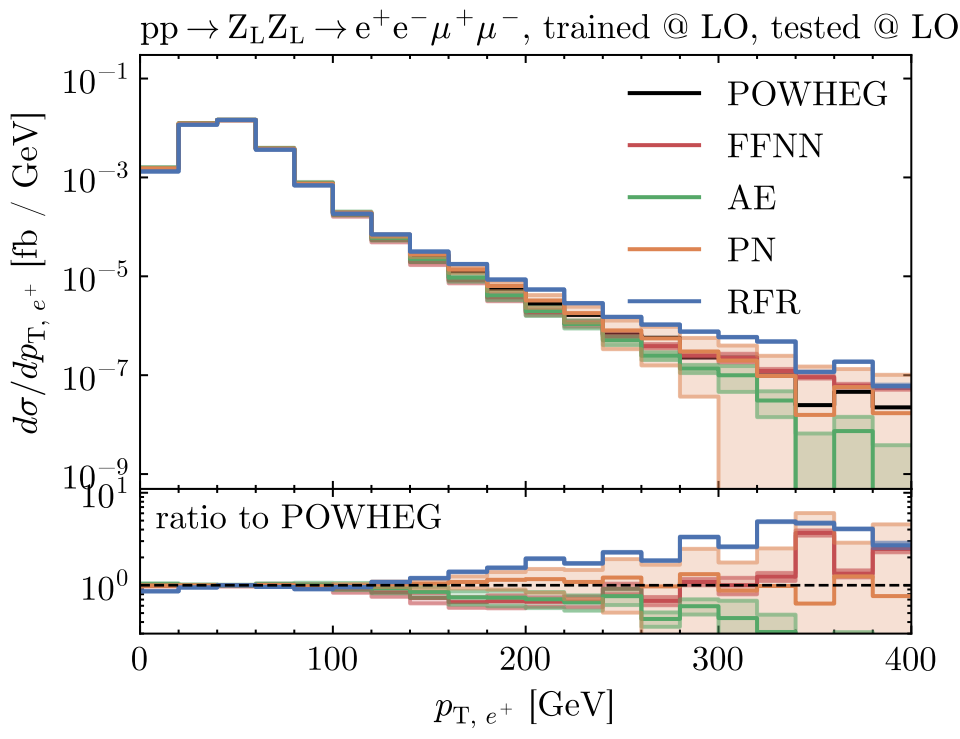
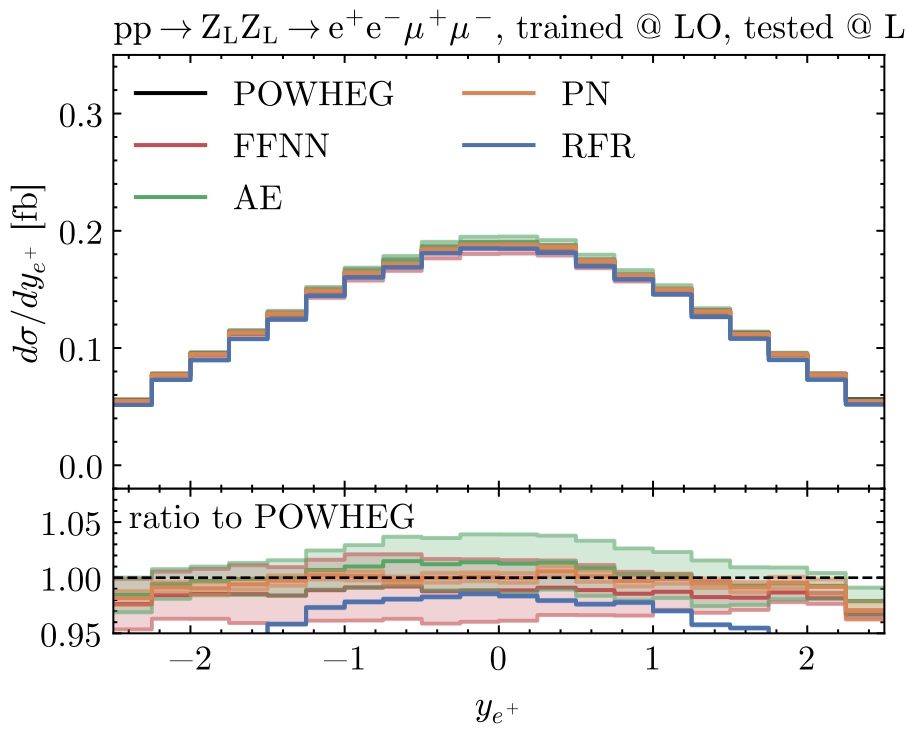
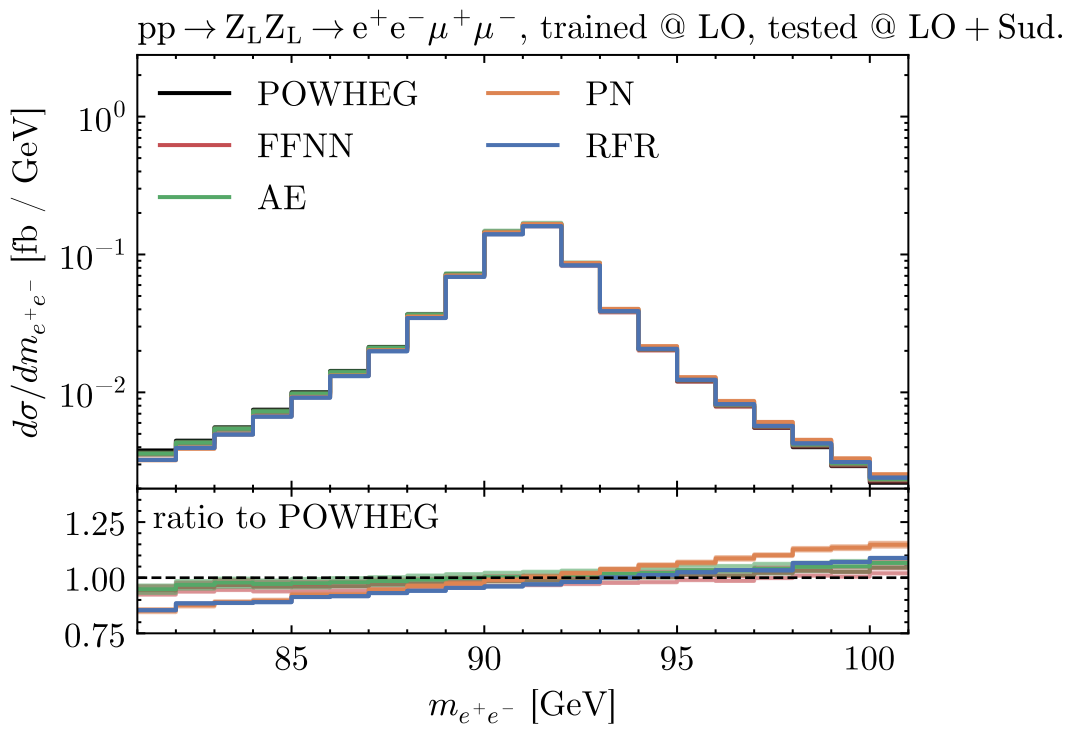
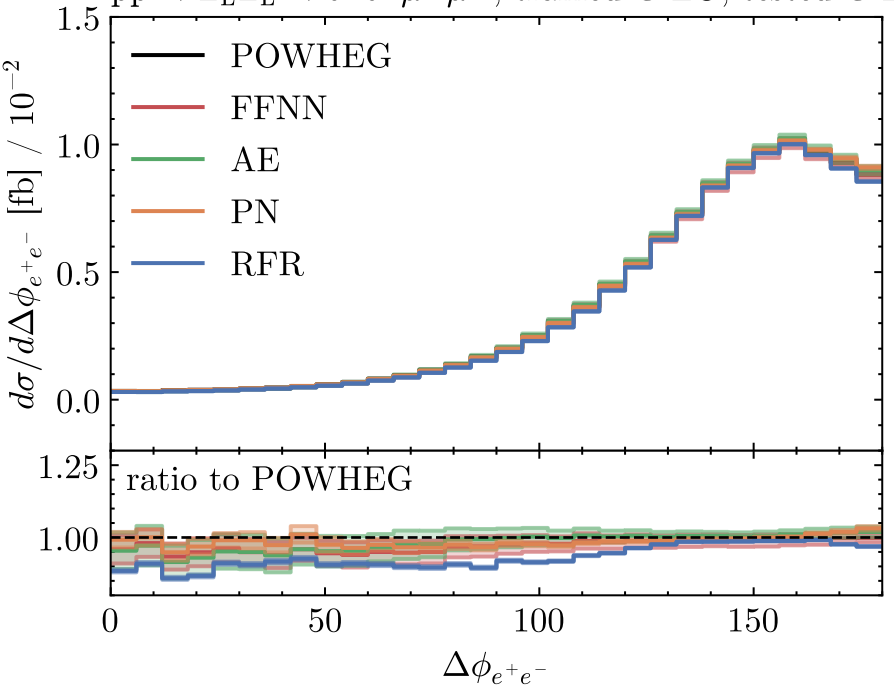
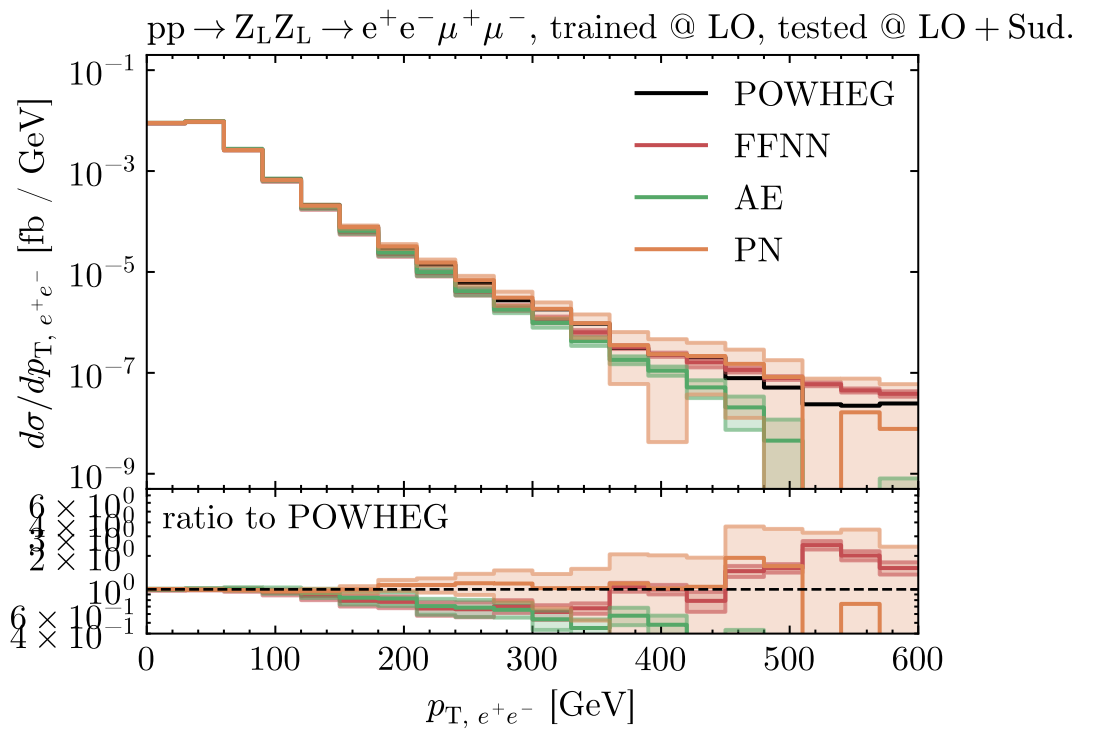

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

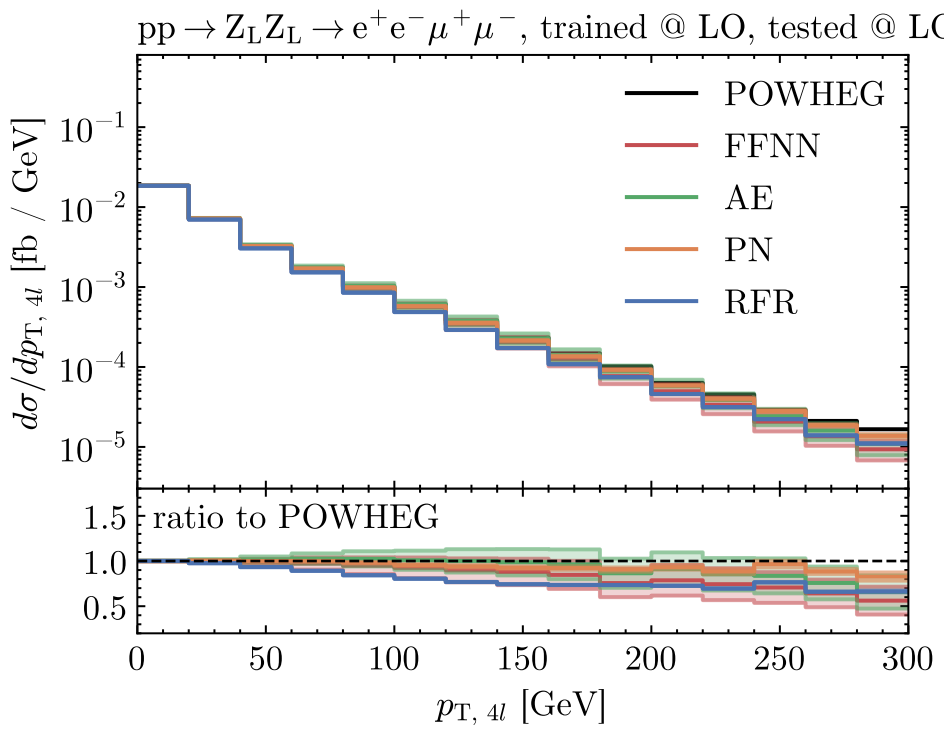

pp → $Z_L Z_L$ → $e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dp_{T,\,e^+}$ [fb / GeV]
ratio to POWHEG
$p_{T,\,e^+}$ [GeV]
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{1}
10^{0}
0
100
200
300
400

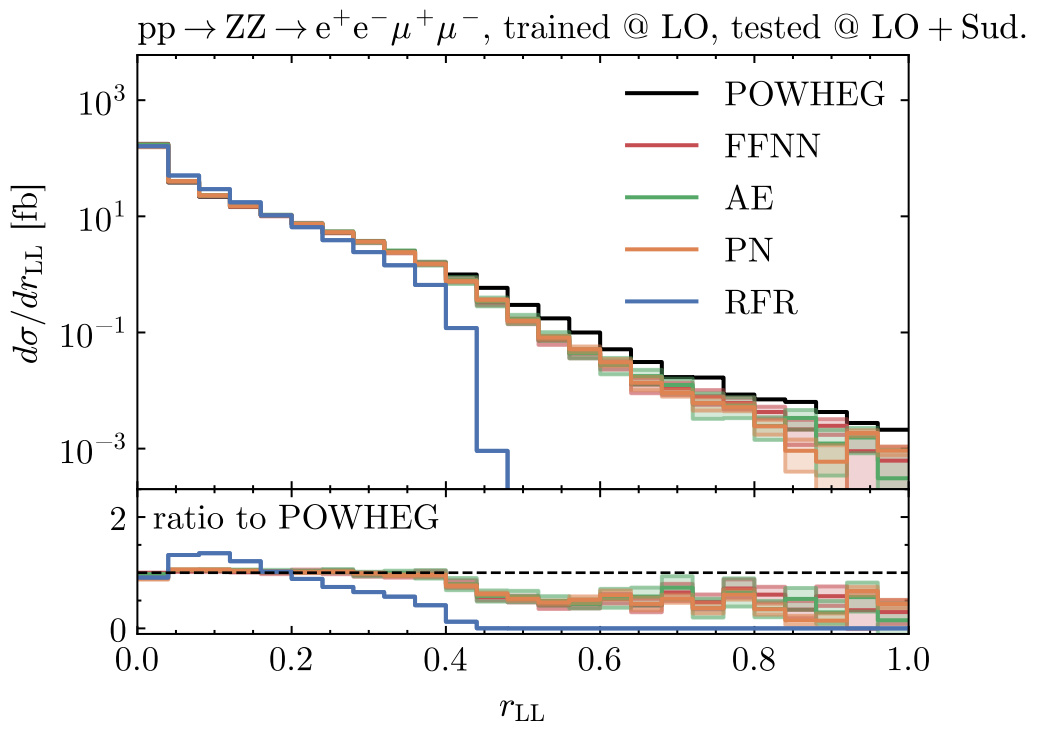

$pp \to Z_L Z_L \to e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dy_{e^+}$ [fb]
0.0
0.1
0.2
0.3
ratio to POWHEG
0.95
1.00
1.05
−2
−1
0
1
2
y_{e^+}


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
dσ/dm_{e⁺e⁻} [fb / GeV]
ratio to POWHEG
m_{e⁺e⁻} [GeV]


$pp \to Z_L Z_L \to e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
$d\sigma/d\Delta\phi_{e^+e^-}$ [fb] / 10^{-2}
1.5
1.0
0.5
0.0
ratio to POWHEG
1.25
1.00
0
50
100
150
$\Delta\phi_{e^+e^-}$


$pp \to Z_L Z_L \to e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/dp_{T,\,e^+e^-}$ [fb / GeV]
ratio to POWHEG
$p_{T,\,e^+e^-}$ [GeV]


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
dσ/dp_T, 4l [fb / GeV]
10^-1
10^-2
10^-3
10^-4
10^-5
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
p_T, 4l [GeV]


pp → ZZ → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dr_{\mathrm{LL}}$ [fb]
10^3
10^1
10^{-1}
10^{-3}
ratio to POWHEG
2
0
0.0
0.2
0.4
0.6
0.8
1.0
r_{LL}